

TUMOR LOCALIZATION · CHIP-DESIGN STUDY

Frequency Reduction

How narrow can the measurement band go before localization degrades?

CNN classification · raw S-parameters · LOSO · measured A3 phantom data

1–5 GHz

recommended band

1/2

the bandwidth of the full sweep

100%

F5 accuracy retained (full array)

Smaller bandwidth, simpler chip

An integrated version of this system lives or dies on front-end complexity: a narrower sweep means simpler synthesizers, antennas and calibration. The question is what accuracy that costs, and where the band should sit.

- 1 WHERE does the information live?** slide a 2 GHz window across 0.1–8 GHz; full 4-antenna array
- 2 HOW WIDE must the band be?** expand 2 → 3 → 4 → 5 GHz around the winning region
- 3 Does it combine with reduced hardware?** winning band × fewer antennas / reflection-only
- 4 Is the band choice robust?** repeat the whole sweep with a single antenna (S11 only)

Every cell: CNN classifier, raw mag+phase input, 100-epoch leave-one-session-out, per-position vote.

A 2 GHz window: the low-mid band wins

Band (GHz)	Width	Empty	A3 + F4	A3 + F5
0.1 – 2	2 GHz	99.4	96.2	90.9
2 – 4	2 GHz	99.4	97.4	92.9
4 – 6	2 GHz	99.4	89.1	71.7
6 – 8	2 GHz	99.4	97.4	69.7
0.1 – 8 (full)	7.9 GHz	99.3	97.4	100.0

All 16 S-parameters. LOSO per-position vote accuracy (%). Highlighted row = best 2 GHz window (mean across phantoms).

The 2-4 GHz window wins. Above 4 GHz, F5 collapses to ~70%: the large lossy glandular insert attenuates high-frequency energy before it reaches interior positions. The empty phantom is band-agnostic.

Expanding around 2-4 GHz: the knee is at 4 GHz

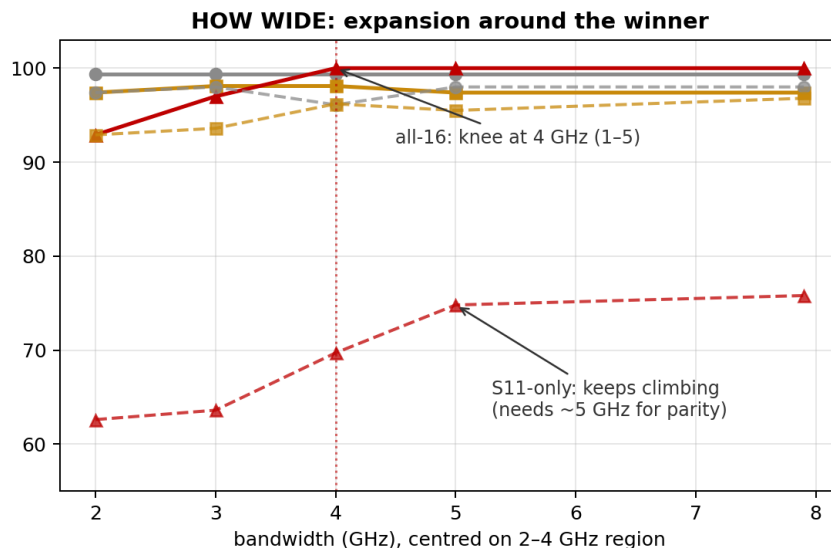
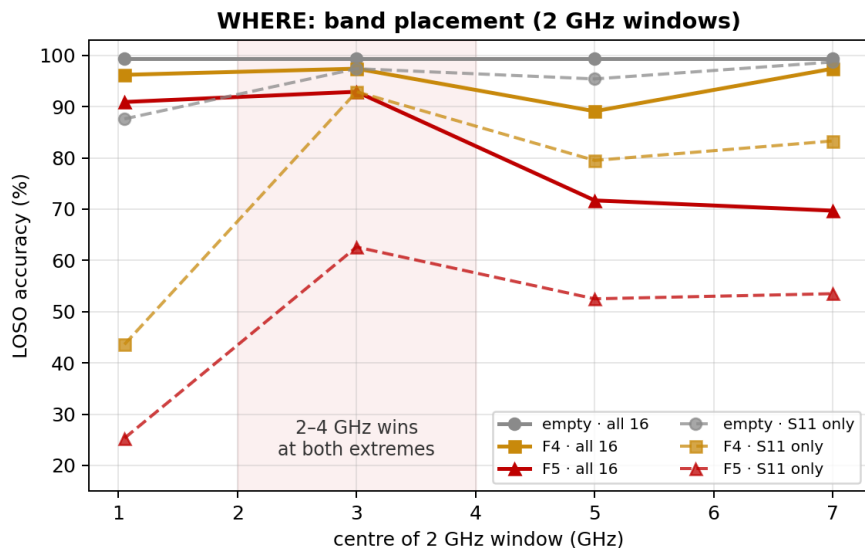
Band (GHz)	Width	Empty	A3 + F4	A3 + F5
2 – 4	2 GHz	99.4	97.4	92.9
1.5 – 4.5	3 GHz	99.4	98.1	97.0
1 – 5	4 GHz	99.4	98.1	100.0
0.5 – 5.5	5 GHz	99.4	97.4	100.0
0.1 – 8 (full)	7.9 GHz	99.3	97.4	100.0

All 16 S-parameters. Windows centred on the winning 2-4 GHz region.

1-5 GHz (4 GHz bandwidth, 3 GHz centre) recovers full-band accuracy on every phantom, including 100% on F5. Going wider adds nothing; going narrower costs F5 3-7 points. Half the bandwidth, zero accuracy cost.

Placement and width, at both hardware extremes

CNN classification vs frequency band: full array (solid) vs single antenna (dashed)



Solid = all 16 S-parameters · dashed = single antenna (S11 only). Left: where the 2 GHz window sits. Right: accuracy vs bandwidth around 2-4 GHz.

STEP 3 · COMBINED REDUCTIONS

The 1-5 GHz band with reduced hardware

Hardware @ 1-5 GHz	Empty	A3 + F4	A3 + F5	F5 @ full band
4 antennas, reflection only	100.0	97.4	94.9	100.0
2 antennas (1 & 3), full S	99.3	92.9	92.9	96.0
2 antennas, reflection only	99.3	92.9	84.8	90.9
1 antenna (S11)	96.1	96.2	69.7	75.8

The reductions compound on F5: a band cut that is free with the full array costs 3-6 points once antennas or transmission paths are also removed. Best reduced operating point: 4 antennas, reflection-only, 1-5 GHz = 95% on the hardest phantom.

STEP 4 · ROBUSTNESS CHECK

The same sweep with one antenna confirms the band

Band (GHz)	Width	Empty	A3 + F4	A3 + F5
0.1 – 2	2 GHz	87.6	43.6	25.3
2 – 4	2 GHz	97.4	92.9	62.6
4 – 6	2 GHz	95.4	79.5	52.5
6 – 8	2 GHz	98.7	83.3	53.5
0.5 – 5.5	5 GHz	98.0	95.5	74.8
0.1 – 8 (full)	7.9 GHz	98.0	96.8	75.8

S11 only. Note 0.1-2 GHz: fine with the full array (90.9 on F5), catastrophic alone (25.3).

2-4 GHz wins again, so the band choice is robust; but with one antenna there is no knee - accuracy climbs to ~5 GHz. Antennas and bandwidth are partially interchangeable information budgets.

Chip band specification

Spec: 1-5 GHz

4 GHz bandwidth, 3 GHz centre. Full-band accuracy on every phantom with the array; the anchor for any hardware configuration.

Fallback: 1.5-4.5 GHz

3 GHz bandwidth costs only 1-3 points on the hardest phantom; 2-4 GHz is the floor (-7 on F5).

Avoid: above 5 GHz and below 1 GHz alone

High bands carry no F5 information (insert attenuation). Pure low band collapses without array diversity.

Design principle

Antennas and bandwidth trade against each other. Cut one aggressively, not both: 4-ant reflection-only @ 1-5 GHz keeps 95% on F5.